

ROBEL FEKADU HAILU

Senior Software Engineer · Backend & AI Systems · Full-Stack

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PROFESSIONAL SUMMARY

Senior software engineer with 5+ years of production experience architecting and shipping scalable backend systems, AI-powered integrations, and cloud-native platforms. Core expertise in Node.js (NestJS), Python (FastAPI, Django), TypeScript, PostgreSQL, and AWS serverless infrastructure. Proven track record delivering complex, high-stakes systems across healthcare, government, fintech, and SaaS domains.

Comfortable owning the full technical lifecycle: architecture, implementation, DevOps, and documentation. Experienced working in distributed, cross-functional teams with real stakeholder accountability.

PROFESSIONAL EXPERIENCE

Senior Engineer (Contract) | [Lyfe / Unvrs](#)

Nov 2025 – Present

- Architected and built Unvrs an AI-assisted home blood sample collection platform with a Next.js 14 (App Router) PWA frontend and a HIPAA-aligned, event-driven serverless backend on AWS (Lambda, API Gateway, DynamoDB, S3, Cognito, EventBridge).
- Defined all Infrastructure as Code using AWS CDK (TypeScript): DynamoDB single-table design, versioned S3 buckets with lifecycle rules, Cognito user pools, API Gateway configuration, and least-privilege IAM policies for Lambda functions.
- Implemented idempotent APIs using idempotency keys and content hashes, along with append-only audit patterns to support safe retries and compliance; integrated device-session and QR-based activation flows with the Lyfe lab-partner backend.
- Built and maintained the Lyfe test-kit marketplace backend (FastAPI, PostgreSQL, Redis, Elasticsearch): order management, kit fulfillment, lab integrations (Pathcare, Synlab, Eurofins), and result processing pipelines; authored deployment runbooks and rollback procedures.
- Maintained deployment automation: AWS CDK deploy scripts (staging and production environments), GitHub Actions CI pipelines (build, lint, test), and ECR/ECS deployment scripts for the FastAPI backend.
- Delivered the Viya client instance (e.g., Viya Kenya) with instance-specific branding, custom domain, and transactional email templates; contributed to the multi-instance architecture enabling fully isolated deployments per client across branding, payment providers, and lab integrations.

Senior Full-Stack Developer | [Mereb Technologies \(LeanHealth\)](#)

Apr 2025 – May-2026

- Led the customization, implementation, and deployment of LeanHealth, a comprehensive EMR system integrated with an ERP platform and OpenMRS covering end-to-end hospital workflows: patient registration, triage, appointments, pharmacy, billing, and lab results.
- Integrated an OpenAI-powered clinical AI chatbot directly into the live EMR system, enabling AI-assisted clinical workflows and decision support the first production AI integration of its kind in the region.
- Developed custom ERP modules to support automated quotation generation, accurate sales attribution, and local payment gateway integration (Santim Pay) via a Node.js SDK.
- Implemented SMS-based patient notifications for lab results through a custom microservice integrated with live EMR data flows.
- Designed and executed data migration pipelines from legacy SQL Server and PostgreSQL databases, mapping historical clinical and patient records to the new system's import formats.
- Architected and maintained Docker-based infrastructure with Nginx reverse proxy, SSL/TLS termination, and environment-specific service profiles for development, staging, and production.
- Built custom analytical reports and operational dashboards, improving visibility for clinical and administrative teams; implemented CI/CD pipelines for automated deployments.
- Collaborated with clinical and cross-functional teams to automate manual processes and improve patient throughput.

Senior Software Engineering Consultant | [Amref Health Africa](#)

Dec 2024 – May 2025

- Architected and delivered a Telegram-based multilingual polling and feedback platform deployed across health centers nationwide, enabling real-time qualitative data collection and staff engagement at scale.
- Designed the scalable data collection system with automated polling schedules, manual feedback triggers, real-time notifications, advanced session management, custom response validation, and dynamic question branching to support diverse user needs.
- Enhanced the bot with multilingual support to accommodate regional languages and streamline feedback processes across geographically distributed health workers.
- Engineered a secure NestJS REST API integrated with Google Looker Studio via a custom AppScript community connector, enabling seamless data sync, advanced analytics, interactive dashboards, and automated reporting for stakeholder decision-making.
- Collaborated with cross-functional teams to define KPIs and optimize Looker Studio visualizations for operational and executive stakeholder presentations.

Senior Software Engineer | [360Ground](#)

Oct 2023 – Apr 2025

- Spearheaded the architecture and development of the Ethiopian Stat Bank system in collaboration with Statistics Norway (SSB) and INSA teams, leading the design and implementation of core microservices backend and frontend components for national statistical data management.
- Led the full technical implementation of the Electronic Market Information System (eMIS) for the Ethiopian Statistics Service built multiple dashboards and a mobile application, significantly improving data collection, real-time market reporting, and accessibility at the national level.
- Contributed to the development of the National Business Portal, a comprehensive permit management system using microservices architecture; implemented the administrative dashboard and public portal, streamlining permit applications for construction and business licenses across Ethiopia.
- Implemented real-time interactive gaming functionality for the live TV game show 'MedaShi', successfully supporting three seasons of live broadcast operation with concurrent user engagement.
- Automated data entry and parsing workflows using custom scripts, reducing manual data import time and improving data accuracy across multiple projects.
- Developed core components of the MedaPay React SDK, enabling streamlined and standardized integration of the MedaPay payment gateway for client-facing websites.
- Migrated critical projects from AWS to on-premise infrastructure, reducing cloud-related costs while enhancing data security and system control.
- Improved user experience for the Meda Cinema ticket booking app by migrating from a super-app to a standalone single-app architecture, simplifying maintenance and improving performance.

Backend Developer | [SoftDeveloper](#)

Jun 2023 – Oct 2023

- Built and documented RESTful APIs for 'Suitor,' a bespoke CRM tool supporting an Australian suit rental company's operations enhancing customer engagement through automated communication workflows and streamlined order management.
- Designed and implemented batch ordering and automated report generation features, enabling scheduled and on-demand report creation for operational teams.
- Developed SMS messaging, authentication, and location management functionalities within the CRM to improve communication efficiency and operational tracking.
- Enhanced order management features including order tracking, scheduling, and customer workflow automation to reduce manual handling.

Software Engineer | [Meklit Technologies](#)

Apr 2022 – May 2023

- Developed a full-stack RND Match Tracking application for sports analytics, managing tracking and scheduling workflows and integrating automated analysis pipelines increasing task efficiency by 40%.
- Designed system architecture and authored comprehensive technical documentation including Software Requirements Specifications (SRS) and System Architecture Documents (SAD) for project alignment and stakeholder transparency.
- Developed static websites and web applications using HTML, CSS, JavaScript, and WordPress for small business clients, delivering user-friendly and visually polished layouts.

TECHNICAL SKILLS

Languages JavaScript, TypeScript, Python, Java

Backend	Node.js, NestJS, Express, FastAPI, Django
Frontend	React, Next.js, Vue.js, Angular
Databases	PostgreSQL, MongoDB, MySQL, Redis, Elasticsearch, DynamoDB
AI / LLM	OpenAI API, Anthropic API, vector embeddings, RAG pipelines, pgvector
ORM / ODM	Prisma, TypeORM
Cloud & IaC	AWS (Lambda, API Gateway, DynamoDB, S3, Cognito, EventBridge, ECS, ECR), AWS CDK, Firebase, Cloudflare
DevOps	Docker, Nginx, GitHub Actions, CI/CD pipelines
Messaging	RabbitMQ, BullMQ
Observability	Elasticsearch, Kibana, Looker Studio, GCP BigQuery
Other	MinIO, ODK, JIRA, Slack, REST, GraphQL

EDUCATION

B.Sc. Software Engineering · Addis Ababa Science and Technology University, Addis Ababa, Ethiopia 2016 – 2021

LANGUAGES

English (C1 Professional Working Proficiency) · Amharic (Native)